

equation (7.2.7) amounts to dropping equation (7.2.4) and instead solving for the coefficients H_0, H_1 that make $y_t = Y_t$ true and that jointly solve equations (7.2.4) and (7.2.7).

It follows that for this example we can compute an equilibrium by forming the optimal linear regulator problem corresponding to the Bellman equation (7.2.13). The optimal policy function for this problem can be used to form the rational expectations $H(Y)$.⁴

7.3. Recursive competitive equilibrium

The equilibrium concept of the previous section is widely used. Following Prescott and Mehra (1980), it is useful to define the equilibrium concept more generally as a *recursive competitive equilibrium*. Let x be a vector of state variables under the control of a representative agent and let X be the vector of those same variables chosen by “the market.” Let Z be a vector of other state variables chosen by “nature”, that is, determined outside the model. The representative agent’s problem is characterized by the Bellman equation

$$v(x, X, Z) = \max_u \{R(x, X, Z, u) + \beta v(x', X', Z')\} \quad (7.3.1)$$

where $'$ denotes next period’s value, and where the maximization is subject to the restrictions:

$$x' = g(x, X, Z, u) \quad (7.3.2)$$

$$X' = G(X, Z). \quad (7.3.3)$$

$$Z' = \zeta(Z) \quad (7.3.4)$$

Here g describes the impact of the representative agent’s controls u on his state x' ; G and ζ describe his beliefs about the evolution of the aggregate state. The solution of the representative agent’s problem is a decision rule

$$u = h(x, X, Z). \quad (7.3.5)$$

⁴ The method of this section was used by Lucas and Prescott (1971). It uses the connection between equilibrium and Pareto optimality expressed in the fundamental theorems of welfare economics. See Mas-Colell, Whinston, and Green (1995).

To make the representative agent representative, we impose $X = x$, but only “after” we have solved the agent’s decision problem. Substituting equation (7.3.5) and $X = x_t$ into equation (7.3.2) gives the *actual* law of motion

$$X' = G_A(X, Z), \quad (7.3.6)$$

where $G_A(X, Z) \equiv g[X, X, Z, h(X, X, Z)]$. We are now ready to propose a definition:

DEFINITION: A *recursive competitive equilibrium* is a policy function h , an actual aggregate law of motion G_A , and a perceived aggregate law G such that (a) Given G , h solves the representative agent’s optimization problem; and (b) h implies that $G_A = G$.

This equilibrium concept is also sometimes called a *rational expectations equilibrium*. The equilibrium concept makes G an outcome of the analysis. The functions giving the representative agent’s expectations about the aggregate state variables contribute no free parameters and are *outcomes* of the analysis. There are no free parameters that characterize expectations.⁵ In exercise 7.1, you are asked to implement this equilibrium concept.

7.4. Markov perfect equilibrium

It is instructive to consider a dynamic model of duopoly. A market has two firms. Each firm recognizes that its output decision will affect the aggregate output and therefore influence the market price. Thus, we drop the assumption of price-taking behavior.⁶ The one-period return function of firm i is

$$R_{it} = p_t y_{it} - .5d(y_{it+1} - y_{it})^2. \quad (7.4.1)$$

There is a demand curve

$$p_t = A_0 - A_1(y_{1t} + y_{2t}). \quad (7.4.2)$$

⁵ This is the sense in which rational expectations models make expectations disappear from a model.

⁶ One consequence of departing from the price-taking framework is that the market outcome will no longer maximize welfare, measured as the sum of consumer and producer surplus. See exercise 7.4 for the case of a monopoly.